

HTPD Abstract

Title: Commissioning a high-repetition rate, 1 J Nd:YAG laser upgrade for HSX Thomson scattering system using multi-passband polychromator filters.

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Thomson scattering (TS) diagnostics are ubiquitous among high-temperature plasma experiments because they provide electron temperature and density measurements with excellent spatial resolution and well-defined uncertainty characterization. TS signal levels are intrinsically low (scattered power fractions on the order of $1\text{E-}14$), so high-energy nanosecond laser pulses and photon-statistics-limited analysis are required. In many systems, the laser repetition rate is a primary limiter on temporal coverage within a discharge and on the ability to reduce random uncertainties through averaging.

We report on the installation and commissioning of a major TS upgrade at the HSX stellarator, centered on a new Nd:YAG laser capable of delivering 1 J, 10-nanosecond pulses at 400 Hz or 2 kHz. The upgrade includes integrating the laser with HSX beam-delivery optics and shot-to-shot monitoring to verify pulse-energy stability. To accommodate the increased data rate, the diagnostic is being coupled to previously fabricated readout electronics optimized for high-repetition operation [1].

In parallel, we use interference filters with multiple spectral passbands in the polychromator to improve spectral utilization and reduce inferred TS uncertainties without increasing detector count [2]. A controlled, multi-objective, elitist genetic algorithm is used to determine optimal multi-passband filter sets for a given channel budget and measurement objectives. For HSX, this enables a compact configuration where a system with only three multi-passband filters has equivalent utility to a standard five-channel system, while maintaining or improving uncertainty relative to a traditional single-passband filter set.

Commissioning results and first TS measurements from the upgraded system will be presented, with emphasis on the laser installation, high-repetition-rate operation, and resulting improvements in temporal resolution and measurement uncertainty compared with the prior configuration.

[1] W. R. Goodman et al, Rev. Sci. Instrum. 93, 093518 (2022)

[2] W. R. Goodman, B. Geiger, Rev. Sci. Instrum. 95, 093507 (2024)